

MUHAMMAD ZAIN-UL-ABIDEEN

AI Full Stack Engineer

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Summary

AI Full Stack Engineer shipping production LLM applications: retrieval-augmented generation (RAG), multi-agent pipelines, and eval harnesses on Python/FastAPI backends with Next.js frontends, backed by automated tests, CI/CD, and observability.

Education

National University of Sciences and Technology (NUST)

Sep 2022 – Jun 2026

Bachelor of Electrical Engineering

Islamabad, Pakistan

Technical Skills

LLM & Agentic AI: RAG, Hybrid Search (BM25, RRF), Multi-Agent Systems, LLM Evaluation (LLM-as-Judge), Prompt Engineering, LangChain, Gemini API, Groq, Ollama, Sentence Transformers, FAISS, ChromaDB, Langfuse

Machine Learning: Supervised & Unsupervised Learning, Ensemble Methods, Feature Engineering, Scikit-learn, PyTorch, TensorFlow, Keras, NumPy, Pandas

Full Stack: Python, FastAPI, REST APIs, JavaScript, Next.js, HTML/CSS, MERN (MongoDB, Express.js, React, Node.js), SQL, SQLAlchemy, SQLite

DevOps & Cloud: Docker, GitHub Actions, CI/CD, Unit & Integration Testing, Git, Vercel, Render, Microsoft Azure, Google Cloud, Streamlit

Projects

Parity – AI Web Accessibility Auditor | Python, FastAPI, Playwright, Gemini parity-iota-puce.vercel.app

- Built a WCAG 2.2 auditor that runs axe-core in headless Chromium (desktop + mobile), adding a pixel-based contrast meter, a keyboard-navigation agent, and a Gemini vision agent that rewrites faulty alt text.
- Raised recall on a 24-issue labelled benchmark from 54% (rules alone) to 24/24 with zero false alarms; consistency checks lifted vision-agent precision from 80% to 100%.
- Shipped FastAPI (Docker, Render) with a job queue, rate limits, and SSRF protection, plus hybrid RAG (BM25 + bge-small, RRF) over 1,092 W3C passages with verified quotes; 190+ tests, eval gates in CI.

Rehnuma – AI Copilot for Electricity Bills | Python, FastAPI, Next.js, Gemini rehnuma-kappa.vercel.app

- Built an Urdu/English copilot that audits electricity bills with a deterministic engine and reads bill photos via Gemini with a self-verifying re-read loop (96.2% field accuracy); forecasts 12 months of bills and prices 2026 solar rules on real meter data.
- Developed hybrid RAG (BM25 + character n-grams, RRF, LLM query rewriting) over 566 NEPRA clauses with a deterministic citation critic; held-out evals: 100% answered, 91% top-5 retrieval, 6/6 out-of-scope queries refused.
- Shipped a FastAPI backend (Docker, Render) and Next.js frontend (Vercel) with 319 tests, 20 eval metrics gated in GitHub Actions CI, Langfuse tracing with PII masking, and a scheduled watcher for NEPRA document changes.

EdgarIQ – SEC Filings Copilot | Python, SEC EDGAR API, Ollama, Groq, JavaScript edgar-iq-web.vercel.app

- Built a grounded multi-agent RAG system over live SEC EDGAR filings with a planner–retriever–drafter–critic pipeline and a numeric grounding checker that verifies every generated figure against retrieved source text.
- Diagnosed a pure semantic-search failure causing wrong-quarter retrieval across near-identical filing periods, then designed a hybrid metadata-filtered retrieval fix that raised golden-set accuracy from 29% to 100%.
- Engineered an automated eval harness (deterministic + LLM-as-judge grading, regression tracking, HTML reports) backed by 89 unit/integration tests that caught real bugs pre-deployment; shipped a live-chat frontend on Vercel.

Experience

Machine Learning Engineer Intern

Jun 2025 – Aug 2025

RISETech

Islamabad, Pakistan

- Collected field data on passing vehicles with an underground vibration, magnetic, and acoustic sensor node.
- Led feature extraction and engineering on the raw sensor signals, building the feature set used for supervised classification into three classes: LTV, HTV, and No Vehicle.
- Trained and evaluated ensemble classifiers (Random Forest, SVM, KNN, LDA, Extra Trees) with Scikit-learn, improving classification accuracy from 71% to 99%.

AI Engineer Intern

Jun 2025 – Aug 2025

Software Productivity Strategists (SPS), NSTP

Islamabad, Pakistan

- Developed backend features for two flagship products: Business Management System (BMS) and Cognitive Service Management (CSM).
- Deployed AI solutions on Microsoft Azure and improved analytics and reporting for a security management system.

Certifications

- Google AI Professional Certificate (Google) • Advanced ML on Google Cloud Specialization (Google Cloud)
- IBM Machine Learning Professional Certificate (IBM) • AWS Cloud Practitioner Essentials (AWS)